



JourneyOS

Your AI Travel Operating System

INTRODUCTION

JourneyOS turns one sentence into a complete travel plan. The user types or speaks what they want — “Plan a 7-day London vacation for me, my wife and two kids” — and an AI assistant gathers only what it genuinely needs, then builds the whole trip: a day-by-day itinerary, flights, hotels, a budget, the passport and visa requirements, and a packing list.

The conversation sits on the left of the screen and a living trip document on the right. Change anything by simply saying so, and only the affected parts are rebuilt. Every price, opening time and entry rule comes from a real data lookup, never from the model's imagination, and the finished plan downloads as a PDF.

PROBLEM STATEMENT

Planning a trip today means assembling it yourself from a dozen disconnected websites.

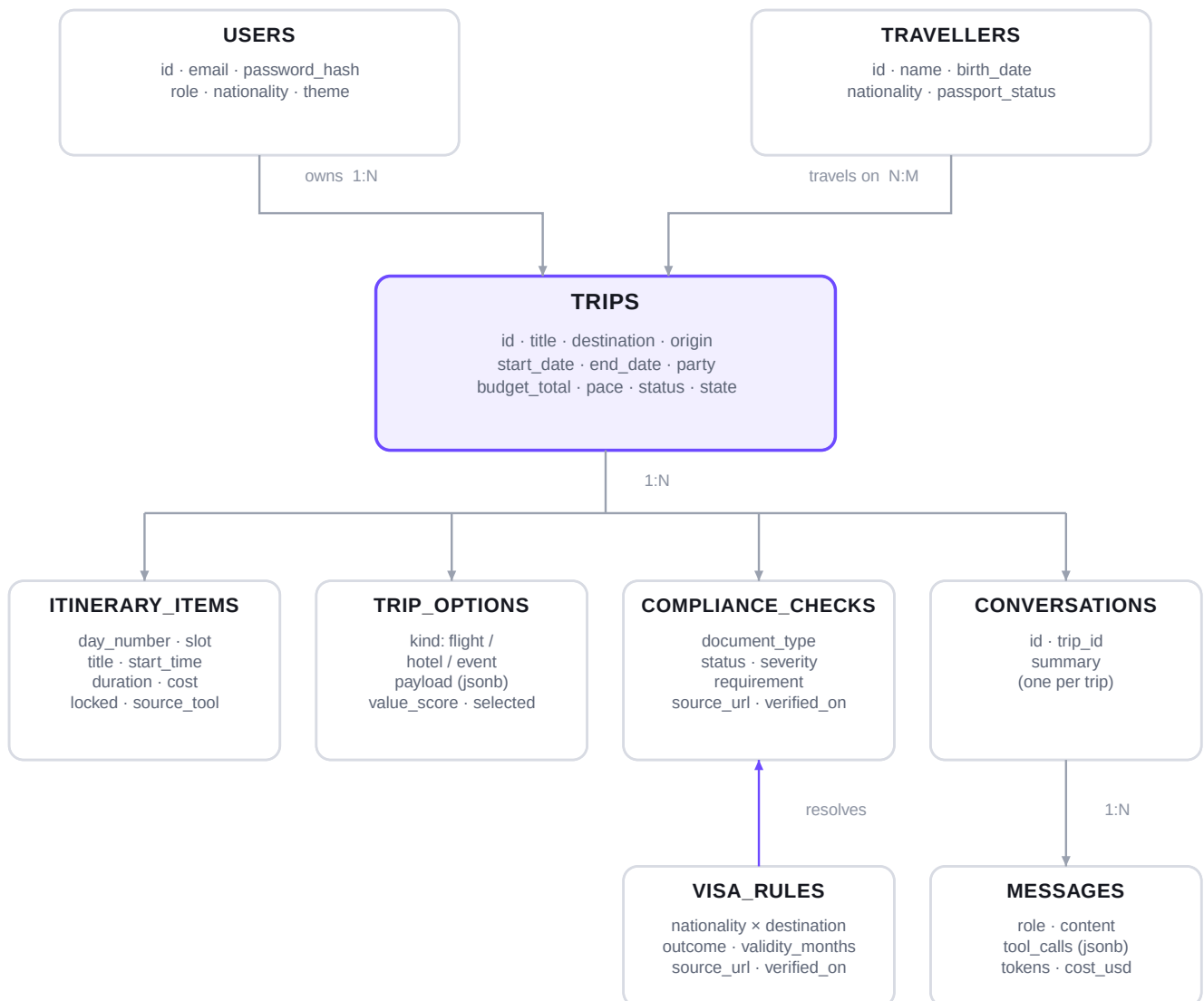
A traveller searches flights on one site, hotels on another, then attractions, events, restaurants, local transport, visa rules, passport validity, currency rates, weather and insurance — each in its own tab, none aware of the others. Nothing shares context, so a single change of dates invalidates half the research and the work starts again.

The consequences are predictable. Families spend twenty hours or more across several weekends. Budgets drift because no tool adds up the whole trip. Itineraries turn out to be physically impossible — museums scheduled while closed, two hours of travel between back-to-back bookings, a six-year-old walked ten kilometres a day. Worst of all, document problems surface too late: a passport inside six months of expiry, or a visa that needed eight weeks of lead time.

No single platform understands what the traveller is actually asking for and manages the trip from intent to departure. That gap is what JourneyOS fills.

DATA MODEL MAP

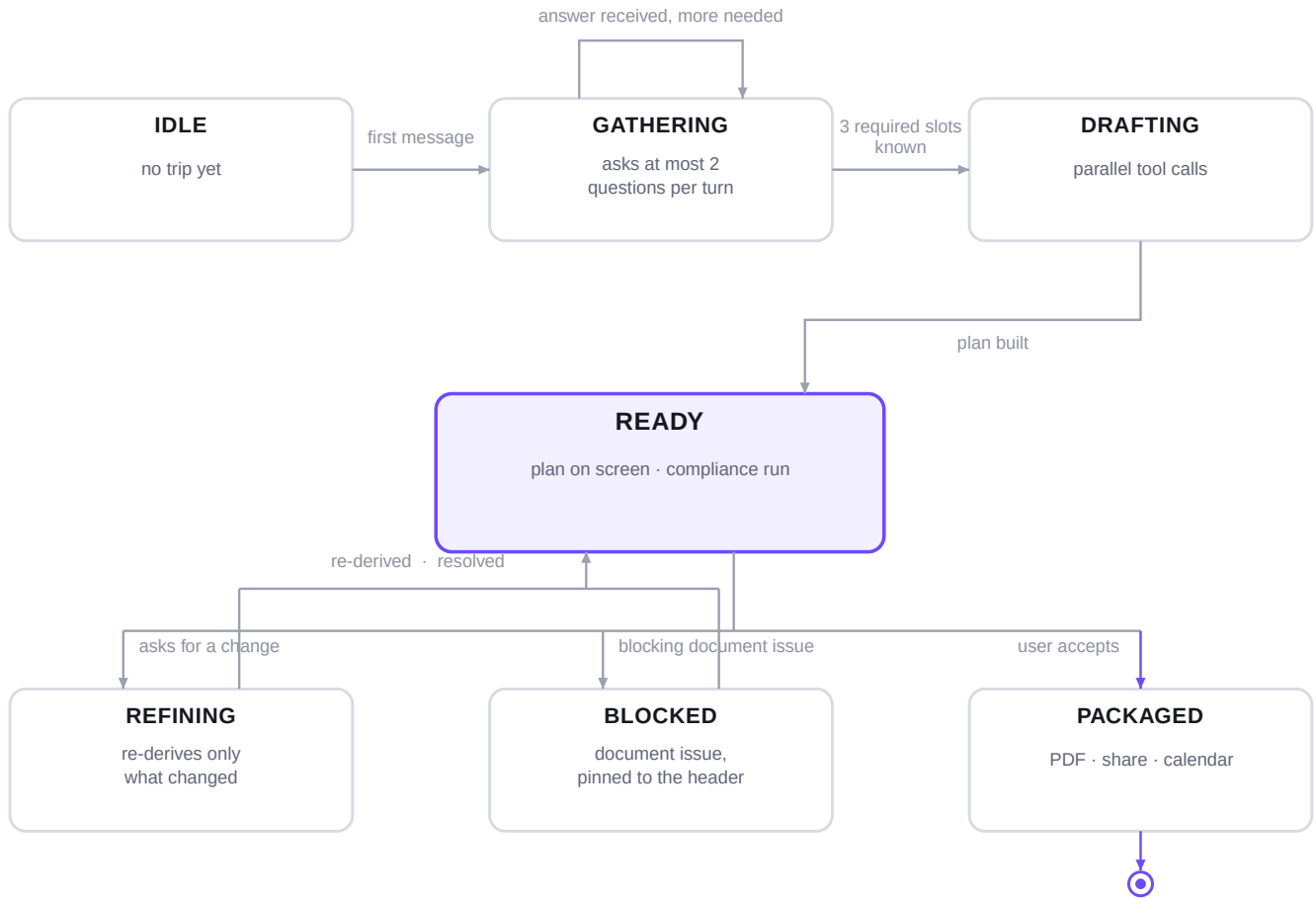
Ten tables in PostgreSQL. A user owns trips and a reusable set of travellers; everything else hangs off a trip. Visa rules are reference data, which is what lets a compliance result cite its source.



Read it as: every fact in a plan is traceable. An itinerary item stores the tool that produced it, and a compliance check stores the rule and the date it was verified.

STATE MACHINE

A trip is a state machine. The assistant's behaviour, what the right-hand pane shows and the main call to action are all derived from this state — never inferred ad hoc.



Two rules the machine enforces: a trip can never reach **PACKAGED** without having passed through compliance at least once, and a change re-derives only what it touches — anything the user locked survives.

OBJECTIVES

- 1 Plan a whole trip from one sentence.**
A 7-day, 4-traveller itinerary with flights, stays, budget and documents in under three minutes.
- 2 Ask less, infer more.**
At most two questions per turn, never for a fact already given. “Me, my wife and two kids” means two adults and two children, with no follow-up.
- 3 Catch document blockers before booking.**
Passport validity and visa requirements checked automatically for every traveller, each citing its source and verification date.
- 4 Hold the budget, or say what broke it.**
Costs allocated across six categories against the user's cap; an overrun names the category that caused it and offers two fixes.
- 5 Produce plans that are physically possible.**
No venue scheduled while closed, travel time shown between stops, and walking capped at 5 km a day when a child under ten is travelling.

TECHNOLOGY STACK

Frontend	Next.js 15 (App Router) · React 19 · TypeScript · Tailwind CSS · shadcn/ui · Framer Motion
AI	Claude Opus 5.5 for planning · Claude Haiku 4.5 for slot extraction · tool calling · SSE streaming
Deterministic engines	TypeScript — route optimisation, time-fitting and budget maths. No LLM in these paths
Database	PostgreSQL 17, local (user postgres, empty password, trust auth) · node-postgres · full-text search
Auth	In-house — bcryptjs hashing · jose JWTs in httpOnly cookies · one seeded admin account
Voice	Web Speech API, with a typed fallback everywhere
Data sources	Seeded JSON for flights, hotels and places · Open-Meteo for weather · seeded visa-rules corpus
Hosting	Vercel · GitHub Actions for CI

One deployable Next.js application. Modular inside, a single process outside — the right shape for a one-day build and for the first fifty thousand users.